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Green-Eyed Monster or Green-Eyed Mirage? A New Procedure for Telling When Begrudging Others' Success Is or Is Not Envy

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Begrudging others' success is a hallmark of envy. Understandably, this has made envy researchers keen to discover the variables that prompt people to begrudge successful others. However, not all negative reactions toward successful individuals stem from envy; for instance, one need not invoke the green-eyed monster to explain our desire to see immoral villains fail. While seemingly uncontroversial, this point poses a challenge to a large and growing body of research that has linked envy with (un)deservingness, finding that undeserved success prompts more ill will than deserved success: Are these negative feelings truly driven by envy or by some other emotion? To help resolve this issue, we introduce the *third-party criterion*—a novel method for ruling out false elicitors of envy. This criterion specifies that if a variable makes potential enviers and third parties (not in a position to experience envy) both begrudge someone's success to similar extents, that variable is unlikely to moderate envy specifically. We report eight studies (four in the main text and four in the Supplemental Materials, $N = 1{,}507$) involving participants recruited from online participant pools between 2022 and 2025 in which we use this procedure to probe variables purported to increase feelings of envy. Ultimately, we find that while some well-established variables do not pass the third-party criterion (e.g., deservingness), others do (e.g., audience valuation). Identifying the precise factors that elicit each emotion is a fundamental goal in emotion research. The third-party criterion offers a simple, widely applicable tool for helping meet that goal.

Keywords: envy, emotion, social comparison, deservingness, attribution

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An old story tells of an envious man who stumbled upon a magic lamp that summoned a genie. The genie offered the man a single wish, but there was a catch: Whatever he wished for, his rival would receive twice as much. Frustrated by this dilemma, the man wrestled over what he should do, the thought of his rival gaining double his boon eating away at him. In a sudden moment of insight, the man exclaimed to the genie, "I've got it! Poke out one of my eyes!".

While extreme, this story articulates an interesting (and dark) aspect of human nature: We sometimes begrudge others merely for having more than we do (Schoeck, 1969). Such reactions typify envy, an emotion that "arises when a person lacks another's superior quality, achievement, or possession and either desires it or wishes that the other lacked it" (Parrott & Smith, 1993, p. 908). Frequently characterized as an emotion triggered by unfavorable upward social comparisons (Lange, Weidman, & Crusius, 2018; R. H. Smith & Kim, 2007; van de Ven & Zeelenberg, 2020), envy can provoke significant distress over others' material possessions and accomplishments. It can drive a diverse range of behaviors, from constructive efforts toward self-improvement to destructive actions such as deception, derogation, complaints of unfairness, social withdrawal, and spreading rumors. Extensive research demonstrates that envy is experienced widely across various contexts and domains (Cohen-Charash & Mueller, 2007; Dunn & Schweitzer, 2006; Moran & Schweitzer, 2008; Parrott & Smith, 1993).

However, while begrudging another's success may be necessary in envy, it is not sufficient. That is, not all begrudging others' success is driven by envy: We can also wish people ill because we dislike them, see them as immoral, or view their gains as ill-gotten. For example, were notorious fraudster Bernie Madoff to succeed in a new financial venture, people might wish him failure and misery (and might even be willing to give up an eye so that he would go blind), but such motivations would likely be driven by moral outrage, hatred, or extreme dislike, and not necessarily by envy. Given that we can wish others ill for a myriad of reasons, how can

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we know whether ill will is driven by envy or some other emotion? While past work has gone to considerable lengths to distinguish envy from related emotions, including jealousy (Parrott & Smith, 1993; R. H. Smith et al., 1988), admiration (van de Ven, 2017; van de Ven et al., 2011), and resentment (Feather, 2008; van de Ven et al., 2011), disagreement remains over envy's precise boundaries and elicitors[1] (Crusius et al., 2020; Lange, Weidman, & Crusius, 2018). For example, van de Ven et al. (2011) have claimed (un)deservingness appraisals elicit malicious envy,[2] while Feather and Nairn (2005) have claimed they instead elicit resentment. Conflation between malicious envy and resentment may obscure important differences between their elicitors, phenomenology, cognition, and behavioral outputs (we expand on this point in our General Discussion section, see e.g., R. H. Smith & Kim, 2007), potentially undermining theoretical precision in understanding how attributions like deservingness influence envy. Indeed, lacking clarity on such issues presents a challenge to the efficacy of emotion research, since understanding cognitive and behavioral processes, predicting decision making, and intervening effectively all often rely on precise theories of underlying emotions (Landers et al., 2024). Is there any way to make empirical progress?

Here, we propose a procedure to help address this challenge. In short, this procedure compares the ill will of people in position to envy a more successful person ("potential enviers") to that of third parties not in a position to envy the same successful person: If potential enviers and third parties both feel ill will toward the more advantaged person to similar extents (as in the case of Madoff), then envy is not likely the cause. In our studies, we demonstrate how this procedure can be used to probe several factors that have been argued to elicit envy, finding that some do not pass muster, while others do. Before getting there, we briefly review past and recent work on envy.

Background

Considered a competitive emotion for monitoring threats to social status (see e.g., Bolló et al., 2020; Crusius & Lange, 2017), envy—known as one of the seven deadly sins—has been observed across times and cultures (Schoeck, 1969; Teitelbaum, 1976; Walcot, 1978), with both ancient (e.g., Aristotle) and modern (e.g., Bertrand Russel) philosophers noting its tendency to prompt negative feelings and actions. Envy also often leads to self-destructive behavior (Thernstrom, 1998; Zizzo & Oswald, 2001). Veblen (1899), for example, famously detailed the role played by envy ("invidious comparison") in "conspicuous consumption," motivating costly and otherwise unnecessary purchases as a signal "of relative ability to pay" (Veblen, 1899, pp. 154–155), furthering what artist Authur R. Momand famously called "Keeping up with the Joneses." Modern economists have since noted the negative consequences for health and well-being that can stem from the envy-driven rat race for status (e.g., Frank, 2001).

Furthermore, as R. H. Smith and Kim (2007) have noted, research has found those in the grip of envy experience joy ("schadenfreude") when envied targets suffer (Leach et al., 2003; R. H. Smith et al., 1996; R. H. Smith & Kim, 2007). Envious individuals can also engage in counterproductive behaviors at work (Braun et al., 2018; A. K. Khan et al., 2014; for review, see Duffy et al., 2021), pay costs to harm others (Parks et al., 2002; Thernstrom, 1998; Zizzo & Oswald, 2001), and destroy useful things rather than see envied others have them (Schimmel, 1993). In extreme cases, envy has even been implicated in individual and intergroup conflict and prejudice (Beck, 1999; Glick, 2002), radicalization (Moncrieff & Lienard, 2024), and violence (de la Mora, 1987; Schoeck, 1969).

However, not all outcomes of envy are similarly dastardly. Research has shown that in addition to hostile feelings and actions, envy can at times produce nonhostile or even constructive reactions (Kaminger et al., 2023; M. H. Khan & Noor, 2020; Parrott & Smith, 1993; Tai et al., 2012), such as improved performance at work (Cohen-Charash, 2009; Crusius et al., 2020; Lee & Duffy, 2019; Schaubroeck & Lam, 2004), inspiration (Kaminger et al., 2023; Meier & Schäfer, 2018), increased effort (A. K. Khan et al., 2017), goal-setting (Lange & Crusius, 2015a), and intentions to help those of higher status (Crusius et al., 2020). Thus, it seems there are cases in which perceiving others outperforming us does not generate such negative feelings. What then determines whether high achievement will elicit negative feelings in others?

Appraisal Models of Envy

To answer this question, many envy researchers point to appraisals—that is, to differences in how the envious individual appraises the situation in which he finds himself and the target of his upward social comparison (Crusius et al., 2020; Heider, 1958; Lazarus, 1991; Salovey & Rodin, 1984; Tesser, 1991). According to one prominent view, envy is composed of two distinct forms: benign envy and malicious envy (Crusius & Lange, 2017). Benign envy motivates individuals to increase their own skills, abilities, and efforts in order to catch up to the envied individual (Crusius & Mussweiler, 2012; Schaubroeck & Lam, 2004; van de Ven et al., 2012), whereas malicious envy motivates individuals to pull down the targets of envy through denigration, derogation, and even outright aggression (Crusius & Lange, 2017; Falcon, 2015; Lange, Paulhus, & Crusius, 2018; van de Ven, 2016). On this "dual[3]" account of envy, when those with whom we compare ourselves perform better than we do at something that is self-relevant to us (Lazarus, 1991; Salovey & Rodin, 1984; Tesser, 1991), we appraise whether the situation is deserved (e.g., someone receives a promotion instead of you because of hard work) or undeserved[4]

[1] We use the terms "elicitor," "variable," and "moderator" interchangeably to refer to antecedents that give rise to envy.

[2] Some researchers consider malicious envy a uniquely hostile subtype of envy, as distinguished from the more constructive benign envy (Lange & Crusius, 2015a; van de Ven, 2016). Throughout this article, we focus exclusively on malicious envy, while other related emotional reactions (e.g., benign envy and resentment) are discussed only where necessary for contrast.

[3] This dual view of envy has established itself within the literature as a highly prominent account (for review, see Lange, Weidman, & Crusius, 2018), and one of its key predictions is that appraisals like deservingness are crucial moderators of envy (Crusius et al., 2020). A cursory search for "malicious envy" on Google Scholar—a proxy for the dual view and its offshoots' (see Lange, Weidman, & Crusius, 2018) distinction between forms of envy—returns hundreds of papers describing experiments taking the dual view for granted (see, e.g., Belk, 2011; Braun et al., 2018; Brooks et al., 2019; Crusius & Lange, 2014; Dong et al., 2020; Feng et al., 2021; Lange & Crusius, 2015a, 2015b; Lange et al., 2016; Salerno et al., 2019; Sterling et al., 2016; van de Ven et al., 2009, 2011, 2015; Vrabel et al., 2018; Xiang et al., 2019). Given its prominence, here, we focus on the dual view and some of its predictions, but our suggested procedure for evaluating moderators of envy would apply equally well to other approaches to envy.

[4] Of note, those who subscribe to the dual view of envy are not the only, nor the first, scholars to highlight the connection between envy and deservingness (e.g., Ben-Ze'ev, 1990, 1992; Miceli & Castelfranchi, 2007).

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(e.g., someone receives a promotion instead of you because of luck or befriending the boss; e.g., Crusius et al., 2020; van de Ven et al., 2012). The experience of benign or malicious envy turns on that appraisal: If the situation is appraised as deserved, people experience more benign envy (and less malicious), and if it is appraised as undeserved, people experience more malicious envy (and less benign; van de Ven et al., 2012). In support of this view, evidence has shown that people do appear to report greater negative feelings toward advantaged targets when their advantages are undeserved than when they are deserved (van de Ven et al., 2012).

To illustrate the dual view's proposed effect of deservingness attributions on envy, consider the following scenario: A chemist learns that a colleague has just been promoted for a groundbreaking discovery. However, the chemist knows this colleague falsified data and plagiarized ideas from others. Naturally, the chemist might feel intense negativity toward their colleague, wishing the colleague would lose their undeserved success and even feeling tremendous satisfaction were that colleague to fall from grace. At first blush, these feelings seem like classic envy (i.e., wishing ill upon a highly successful social competitor), and indeed many researchers have interpreted such (un)deservingness scenarios as eliciting malicious envy (Ferreira & Botelho, 2021; Ortony et al., 1988; van de Ven et al., 2012). In support of this account, experiments have shown that such (un)deservingness manipulations certainly do augment negative feelings toward an undeserving target's success (van de Ven et al., 2012).

But now consider the reaction of a third-party observer who has heard the same story about the plagiarizing chemist but has no interest in chemistry nor any competitive inclination toward the chemist. If this third-party[5] observer also experiences negative feelings toward the plagiarizing chemist (which we suspect they might), would we want to call this reaction envy? What would their feeling "envy" in such a situation mean? Our hypothetical third-party does not know the parties involved in the scenario, and another's success or failure in chemistry—deserved or undeserved—means nothing to them. Indeed, a third-party's response to such a scenario would likely parallel how they would feel were they asked about the villain in the latest superhero movie or a thief who stole from the needy (Hoorn & Konijn, 2003)—they would want to see him get his just deserts.

Moreover, we would not want to call such a third-party reaction envy because researchers have consistently recognized envy as fundamentally rooted in processes of social comparison, where individuals assess their own status, abilities, or possessions relative to others (Festinger, 1954; R. H. Smith & Kim, 2007). Social-comparison theory describes how comparing ourselves with others is a fundamental cognitive process that shapes self-perceptions and emotional reactions, often generating negative affect when the comparison highlights one's inferiority. And many envy theorists emphasize that envy emerges directly from these very comparative processes (Lazarus, 1991; Salovey & Rodin, 1984; R. H. Smith et al., 1994; R. H. Smith & Kim, 2007; Tesser, 1991; Vecchio, 2000). If envy arises from social comparison, it follows that envy should predominantly occur when one is in a position to make comparative judgments about how a target relates to the self. By this reasoning, it would be odd if a third-party observer's negative feelings toward the plagiarizing chemist constituted envy, since third parties are unlikely to be driven by social comparison or competition with the plagiarizer. Thus, if uninvolved third parties—with no reason to envy—nevertheless harbor these negative feelings, then envy is not necessary to produce them. Instead, these negative feelings likely stem from moral outrage, anger at unfairness, or a desire to see a moral wrongdoer punished (Delton & Krasnow, 2017; Fehr & Fischbacher, 2004; Fincher & Tetlock, 2015; Henrich et al., 2010; Krasnow et al., 2016; McAuliffe et al., 2015; Seip et al., 2014; Yamagishi et al, 2009). As moral psychologists have documented frequently (e.g., Darley, 2009; Greene & Haidt, 2002; Jordan et al., 2016), humans enjoy seeing bad things happen to bad people, and desire to see others—including third parties—rewarded fairly in proportion to what they deserve (e.g., Adams, 1965; Baumard et al., 2013; Shaw et al., 2019; Sigelman, & Waitzman, 1991). Moral or fairness-based concerns sufficiently explain these reactions, no envy required.

The Third-Party Criterion

While perhaps uncontroversial, this distinction between envy on the one hand and nonenvious negative feelings on the other presents a quandary for envy researchers: If third parties' ill will toward immoral or undeserving people is unlikely to be driven by envy, how can we determine whether first parties' ill will toward these same people is motivated by envy in similar situations? As we noted above, moral and fairness-based reactions may be sufficient to explain these feelings (for third parties, but perhaps also for first parties in a position to envy the advantaged target).

Here, we propose a solution to this problem. We start from the premise—as virtually all envy theorists do (e.g., Crusius & Lange, 2017; Lazarus et al., 2001; Parrott & Smith, 1993)—that envy is an interpersonal emotion experienced by those who feel threatened by the success of a competitor. This competitor is typically the target of one's upward social comparison who holds an advantage that matters to the envier's self-worth or ego (van de Ven et al., 2009).

Building on this premise, we propose a straightforward test for isolating whether a given variable specifically moderates envy rather than causing general negative affect toward a successful target: Determining whether the variable in question increases people's negative affect when the evaluator (i.e., the one experiencing the negative affect) is a potential competitor rather than a noncompetitor. This test can be accomplished by comparing first parties' (those in a competitive relationship with the advantaged target and thus in a position to experience envy) and third parties' (those not in a competitive relationship with the target and thus not in a position to experience envy) negative reactions to the advantaged target.

The logic behind this test—what we term the *third-party criterion*—is simple: Any variable hypothesized to specifically

[5] By "third party," we mean any individuals who do not consider themselves directly related to or involved with the envy-inducing situation. Previous research has construed envy as involving a relation between an envier (what we refer to as "first party") and the target of that envier's upward social comparison with respect to an envied object (Foster et al., 1972; Rodriguez Mosquera et al., 2010; R. H. Smith et al., 1999). A third party to this relationship is any individual who either lacks desire for the envied object or who lacks any competitive association with the target of envy (i.e., the envied target exists outside of the third party's domain of self-relevance, see Salovey, 1991; Takahashi et al., 2009) and thus lacks a reason for social comparison with the envied target, feelings of inferiority, or both. While envy toward successful others can occur even outside domains of self-relevance, research has consistently demonstrated that envy toward those who share a domain of self-relevance is much stronger (e.g., Dvash & Shamay-Tsoory, 2011; Lin et al., 2018).

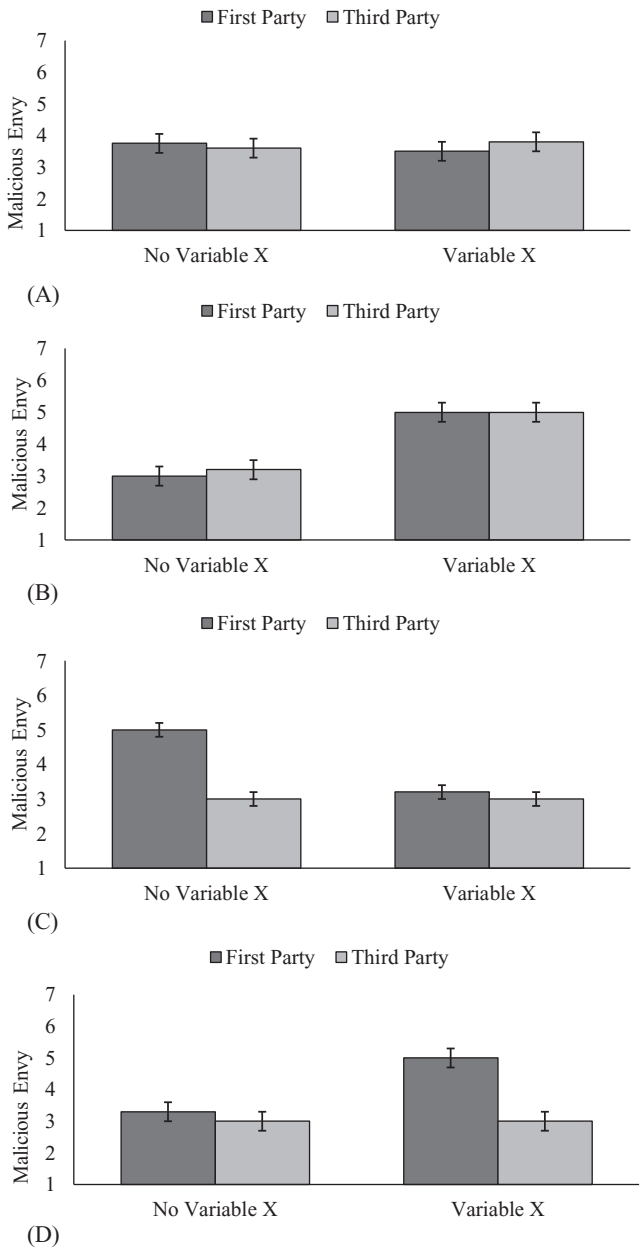
moderate envy should increase negative affect significantly more for first parties than for third parties. Thus, if a variable leads both first parties and third parties to feel similarly negative about a successful target, it likely indicates that the variable in question is influencing general negative affect rather than envy specifically. To demonstrate the effect of a variable on envy, one must show an interaction, such that the variable in question increases first parties' negative reactions to an envied target over and above third parties' reactions to the same target. In other words, to demonstrate a variable's effect on envy, it is not enough to demonstrate the variable elicits greater ill will when it should; one must also demonstrate that the variable does not elicit greater ill will when it should not (for similar logic to this argument in the domain of top-down effects on perception, see Firestone & Scholl, 2016).

When applying the third-party criterion, there are several possible outcomes for any variable purporting to moderate envy: for instance, (a) the variable has no effect on negative feelings, (b) the variable has only a main effect on negative feelings, (c) the variable shows an interaction with party such that the variable decreases negative feelings in first parties more than third parties,[6] and (d) the variable shows an interaction with party such that the variable augments negative feelings in first parties more than third parties. Only outcome (d) satisfies the third-party criterion. For a visual illustration of these four possible outcomes, see Figure 1A–1D.

We propose that the third-party criterion is particularly useful as a tool for ruling out unlikely moderators of envy. If a variable fails this criterion—showing, for example, similar negative reactions among both first and third parties—this suggests the variable may not be specifically manipulating envy at all, but rather a broader set of negative emotions or attitudes. In the studies below, we use the third-party criterion to show that while some established variables in the literature pass this test (audience valuation [Study 2; Supplemental Study S4] and domain of comparison [Supplemental Study S2]), others do not ([un]deservingness [Studies 1a, 1b; Supplemental Studies S1 and S3] and likeability [Study 3]).

Figure 1

Four Possible Outcomes for a Variable Purporting to Moderate Malicious Envy When Applying the Third-Party Criterion



Note. From top to bottom: (A) No effect (fails the third-party criterion), (B) main effect only (fails the third-party criterion), (C) interaction showing reduced effect of variable on first parties (fails third-party criterion), and (D) interaction showing augmented effect of variable on first parties (passes the third-party criterion).

General Methods

Transparency and Openness

For all experiments, informed consent was obtained using a standard survey question. All data were collected from online platforms (CloudResearch and Prolific) between 2022 and 2025. While we collected basic demographic information (age and gender), we did not formulate any specific a priori hypotheses concerning demographic or personality variables. Accordingly, we did not collect additional demographic details such as socioeconomic status, ethnicity, or personality traits, as these were not directly relevant to the primary theoretical aims of our research. Each study employs a single vignette scenario, which we then conceptually replicate across multiple, preregistered independent samples (see Studies 1a and 2; Supplemental Studies S2–S4). All experiments were preregistered before data collection began. All data, code, surveys, and preregistrations are available at https://osf.io/euj3a/?view_only=a31bb007d5b746fc910fe50524700db8 (Landers & Shaw, 2025). All data wrangling and analyses were done using excel, G*Power, and R. All descriptive statistics for individual measures in each study can be found in the additional online materials on the Open Science Framework page (https://osf.io/euj3a/?view_only=a31bb007d5b746fc910fe50524700db8). For all studies (Studies 1–3 in the main text, as well as Supplemental Studies S1–S4), we conducted a priori power analyses based on an anticipated small-to-medium

[6] Such an interaction would especially violate the third-party criterion, because it suggests the variable is *reducing* differences in negative feelings that exist between first and third parties.

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effect size ($\eta_p^2 \approx .02$), which recommended around 50 participants per cell.

Studies 1a and 1b: Deservingness Replications. Does (Un)Deservingness Increase Envy—or Just General Negativity?

In Studies 1a and 1b, we investigate whether manipulations of (un)deservingness pass the third-party criterion. That is, we test whether (un)deservingness specifically increases the negative feelings of first parties or instead causes broader negative reactions that need not be rooted in envy. It seems clear based on previous data and theorizing that people feel much more negative about a target's success when it is undeserved (e.g., van de Ven et al., 2012). The key question is whether these negative feelings are driven by envy or by something else. According to the dual account of envy, manipulations of (un)deservingness increase negative feelings toward the target through increased malicious envy. Therefore, when applying the third-party criterion, the dual account predicts that (un)deservingness will cause a steeper increase in the negative feelings of first parties (who could be experiencing envy) than in the negative feelings of third parties (who should not be experiencing envy).

While obtaining an interaction between party and (un)deservingness certainly seems possible (for instance, maybe the combination of envious feelings and unfairness is a particularly galling brew) and is a clear prediction of the dual account of envy, it also seems highly possible that manipulating (un)deservingness will merely cause a main effect (e.g., [un]deservingness will cause an increase in negative feelings among both first and third parties). Indeed, while envy has been associated with a "subjective sense of injustice" (R. H. Smith & Kim, 2007, p. 49), (un)deservingness seems to prompt intuitions about unfairness that are not necessarily connected to any feeling of envy (Feather & Nairn, 2005; R. H. Smith, 1991; R. H. Smith et al., 1994). This seems particularly true for cases in which the (un)deservingness is clear-cut (as is the case in many manipulations of malicious envy in the literature): In such scenarios, it seems likely for all parties—both involved and uninvolved—to harbor negative feelings toward the target's success. Based on this logic, we hypothesize that we will observe no interaction between party and (un)deservingness, but instead simply a main effect—both involved and uninvolved third parties will wish the undeserving, successful target ill (the pattern represented in Figure 1B). Thus, we predict that (un)deservingness will not pass the third-party criterion, suggesting that past work may have conflated negativity rooted in unfairness (or other nonenvious feelings) with negativity rooted in malicious envy.

Study 1a

In Study 1a, we examine whether (un)deservingness can pass the third-party criterion using a vignette adapted from a well-cited experiment in the literature (van de Ven et al., 2012, cited over 300 times) purporting to demonstrate that appraisals of (un)deservingness serve as an antecedent to malicious envy. Rather than an interaction, we predict that (un)deservingness will instead cause a main effect of increased negative feelings, with no interaction with one's role (first vs. third party). Note, here we directly replicated van de Ven et al.'s exact methods, vignette, and measures (to rule out any concern that we may fail to find evidence that deservingness passes the third-party criterion due to modifying their methods).

Method

We preregistered this study at https://aspredicted.org/SK6_T1M.

Participants

We predetermined to recruit 205 participants from CloudResearch who had 95%+ prior survey approval and were U.S. participants. Participants completed one attention check at the beginning of the survey and were prevented from taking the rest of the survey if they failed this question. At the end of the survey, participants were presented with three comprehension check questions; we planned to exclude participants from analysis if they answered two or more of these questions incorrectly. Of those recruited who completed the survey, 203 participants passed the initial attention check. Of those, 17 failed at least two comprehension checks or a final attention check and were thus excluded from analysis. In total, we analyzed data from 186 participants (89 women, 97 men, M_{age} = 39.17 years, SD = 11.43) who completed the survey in exchange for $0.45. Sensitivity power analysis suggested this sample size provides 80% power to detect an effect of $\eta_p^2 = .02$.

Experiment Design

We assigned participants to one of four conditions in a 2 (party: first, third) × 2 (deservingness: deserved, undeserved) between-participants design.

Procedure

Participants read a vignette adapted from van de Ven et al. (2012) describing a paradigmatic envy-eliciting scenario occurring at work: One employee is promoted and receives a higher salary than a second, target employee. Participants read that each year, the employees at the company are evaluated on their job performance, and their salaries are set based on that evaluation. Across conditions, we varied whether the employee who received the promotion and salary increase performed better (deserved condition) or worse (undeserved condition) on the yearly evaluation than the target employee. In addition, we varied across conditions whether the participant took the role of the target employee (first-party condition) or read about someone else in that position (third-party condition). For example, in the third-party undeserved condition, participants read,

> Bob and some fellow colleagues are working in a call center to earn some extra money. The call center is part of a big office supplier and targets large companies. It is Bob's job to contact existing clients for new orders, and to reel in some new clients. As performance is important for management, everyone is evaluated on a yearly basis, and Bob's wage for the upcoming year is determined based on this evaluation.
>
> Last week the evaluations took place and Bob's new wage is set. There is one thing that frustrates Bob: One of his coworkers, whom Bob often hangs out with and gets along with reasonably well, ends up receiving a higher wage than Bob, *despite the fact that the evaluation actually shows that Bob's coworker reels in less clients and also sells less.*

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In the first-party conditions, the vignette instead described the participant as the target employee in place of Bob. And in the deserved conditions, the final line of the vignette, italicized above, was replaced with, "given the fact that the evaluation shows that his [your] coworker reels in more clients and also sells more."

After reading the vignette, participants answered two questions assessing their own malicious envy toward the target, either as a first party making judgements about their own successful coworker or as a third party making judgements about Bob's successful coworker. These questions were adapted from van de Ven et al. (2012) and aggregated into a single index ($r = .58$, $p < .001$): "Would you secretly wish for your [Bob's] coworker to lose clients?" (1 = *not at all*, 8 = *very much so*) and "Would you gossip about your [Bob's] coworker to others?" (1 = *not at all*, 8 = *very much so*), respectively. Because the correlation between the two malicious envy items was below .6, we also report analyses of each item considered individually in the Supplemental Materials. As a manipulation check, we also measured the perceived deservingness of the situation: "How deserved do you [does Bob] think this situation is?" (1 = *undeserved*, 9 = *deserved*). The manipulation was successful: Participants thought their (or Bob's) coworker deserved the higher raise more when the evaluation showed that he reeled in more clients than the participant (or Bob; $M = 6.02$, $SD = 2.55$) than when the evaluation showed the participant (or Bob) had reeled in more clients than the coworker ($M = 2.75$, $SD = 2.29$), $t(179.86) = 9.20$, $p < .001$, $d = 1.35$. Finally, participants reported their demographic information (age, sex).

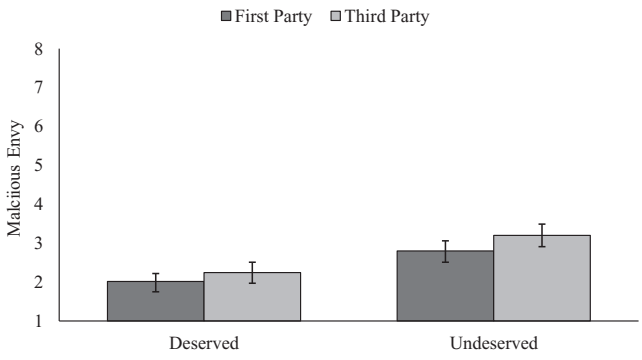
Results and Discussion

Did Deservingness Pass the Third-Party Criterion?

Envy. No—a 2 (party: first, third) × 2 (deservingness: deserved, undeserved) analysis of variance (ANOVA) on target malicious envy ratings revealed, as expected on both accounts, a main effect of deservingness, $F(1, 182) = 11.43$, $p < .001$, $\eta_p^2 = .06$ (see Figure 2), such that participants expressed more "malicious envy" (or "negative reactions to others") when the situation was undeserved ($M = 3.01$, $SD = 1.90$) than when the situation was deserved ($M = 2.13$, $SD = 1.59$), $t(180.80) = 3.42$, $p < .001$, $d = .50$. There was no main effect of party, $F(1, 182) = 1.66$, $p = .200$, $\eta_p^2 < .01$. Most importantly, as we predicted, and counter to accounts positing that manipulations of deservingness moderate malicious envy, there was no interaction between party and deservingness, $F(1, 182) = 0.12$, $p = .732$, $\eta_p^2 < .01$.

Figure 2

Malicious Envy Reported by Participants in Study 1a



Note. In Study 1a, the participants reported malicious envy when the situation was deserved or undeserved and when the participant rated the malicious envy of a first or third party. Malicious envy scale goes from 1 (*low*) to 8 (*high*). Error bars display standard error.

The results of Study 1a provide evidence that deservingness does not pass the third-party criterion: Although there was a very large main effect of deservingness, there was no interaction. That is, (un)deservingness caused participants to attribute negative feelings to both potential enviers (first parties) and nonenviers (third parties) to the same extent. These results suggest that deservingness does not uniquely impact malicious envy, and in fact, there is no way to tell whether it impacts malicious envy at all: People's negative feelings in this situation may entirely be rooted in unfairness and not envy.

Study 1b

To test the generalizability of our results from Study 1a, in Study 1b, we examine whether (un)deservingness can pass the third-party criterion using a vignette adapted from Ferreira and Botelho (2021) in which a business student learns of another student who has achieved the highest grades in the class due to deserved (e.g., hard work) or undeserved (e.g., cheating) reasons. Here, unlike in Study 1a, we did not ask participants to imagine themselves in the place of the business student, but rather asked them to evaluate the envy of another person who is a first party or third party to the envy-inducing scenario. Because we hypothesized that manipulating (un)deservingness would induce general negative reactions rooted in unfairness rather than more specific negative reactions rooted in envy, we again predicted that (un)deservingness would activate negative reactions in both first and third parties, with no interaction between deservingness and party (i.e., deservingness would not pass the third-party criterion).

Method

We preregistered this study at https://aspredicted.org/Z2W_CLC.

Participants

We predetermined to recruit 205 U.S. participants from Prolific. Participants completed one attention check at the beginning of the survey and were prevented from taking the rest of the survey if they failed this question. At the end of the survey, participants were presented with three comprehension check questions; we planned to exclude participants from analysis if they answered two or more of these questions incorrectly. Of the participants recruited who completed the survey, 208 passed the initial attention check. Of those, four failed at least two comprehension checks and were thus excluded from analysis. In total, we analyzed data from 204 participants (106 women, 98 men, $M_{age} = 37.72$ years, $SD = 13.42$) who completed the survey in exchange for $0.65. Sensitivity power analysis suggested this sample size provides 80% power to detect an effect of $\eta_p^2 = .02$.

Experiment Design

We assigned participants to one of four conditions in a 2 (party: first, third) × 2 (deservingness: deserved, undeserved) between-participants design.

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Procedure

Participants read a vignette adapted from Ferreira and Botelho (2021) describing a new paradigmatic envy-eliciting scenario occurring at school: One student (referred to as the "COLLEAGUE") receives the highest grades in the class, a higher grade than a second, target student called Bob. Participants read that students who receive higher grades are better positioned for career success. Across conditions, we varied whether Bob's COLLEAGUE worked hard (deserved condition) or cheated (undeserved condition). Across all conditions, participants then read about a third character, Eugene, who does not know or go to school with either student but has heard about how the COLLEAGUE's cheating (in the undeserved condition) or their hard work (in the deserved condition) led to their receiving a higher grade than Bob. For example, in the undeserved conditions, participants read,

> Imagine Bob is a freshman undergraduate student in the business school of a major university.
>
> Recent research from major universities shows that freshmen business undergraduate students' grades impact their career and success. Students with higher grades at the beginning of their course tend to: (i) be better positioned with a strong knowledge base, and (ii) be in advantage in the selection process for international exchange programs.
>
> Imagine that a classmate of Bob's (we'll use the name COLLEAGUE to designate him/her) got the best grades of the 1st Semester (and Bob didn't). *COLLEAGUE is smart, but studied little for the exams, s/he used dishonest methods, such as cheating*.
>
> Eugene is an accountant who does not go to school and does not know either Bob or Bob's COLLEAGUE. Eugene has heard from a friend about how Bob's COLLEAGUE has cheated to get the best grades in Bob's class.

In the deserved conditions, the line in the vignette italicized above was replaced with, "COLLEAGUE is smart, studied hard for the exams, put a lot of effort with much dedication and motivation."

To manipulate first versus third party, across conditions we varied whether participants were tasked with assessing Bob's malicious envy for his COLLEAGUE (first-party condition) or Eugene's malicious envy for Bob's COLLEAGUE (third-party condition). The questions participants answered were adapted from Ferreira and Botelho (2021), (r = .65, p < .001) and aggregated into a single index: "If this story occurred, how would [Bob/Eugene] feel about the classmate? He would wish that s/he would get these grades" (1 = *he wishes s/he had not gotten these grades*, 5 = *he wishes s/he had gotten these grades*) and "If this story occurred, how would Bob/Eugene feel about [the/Bob's] classmate? He would speak of her/him to others" (1 = *he would speak very badly of her/him to other people*, 5 = *he would speak very well of her/him to other people*), respectively.

As a manipulation check, we also measured the perceived deservingness of the situation: "Does Bob/Eugene think [his/Bob's] colleague deserved to have the best grades?" (1 = *they do not deserve at all*, 5 = *they deserve a lot*). The manipulation was successful: Participants thought Bob or Eugene believed Bob's colleague was more deserving of the highest grades when the colleague studied hard (M = 4.26, SD = .79) than when the colleague cheated (M = 1.53, SD = .76), $t(202)$ = 25.21, p < .001, d = 3.53. Finally, participants reported their demographic information (age, sex).

Results and Discussion

Did Deservingness Pass the Third-Party Criterion?

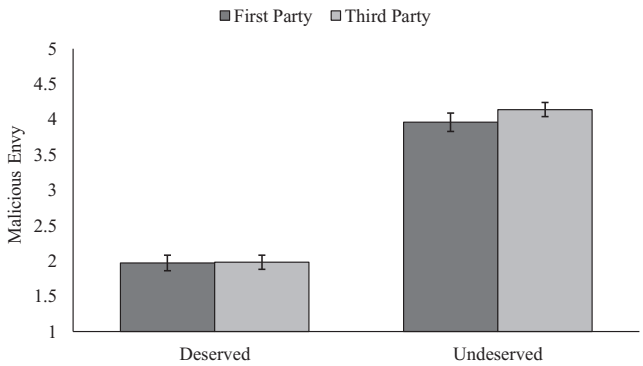
Envy. No—as predicted, a 2 (party: first, third) × 2 (deservingness: deserved, undeserved) ANOVA on target malicious envy ratings revealed only a main effect of deservingness, $F(1, 200)$ = 343.95, p < .001, η_p^2 = .63 (see Figure 3), such that participants evaluated others as more maliciously envious when the situation was undeserved (M = 4.05, SD = .83) than when the situation was deserved (M = 1.98, SD = .76), $t(198.67)$ = 18.54, p < .001, d = 2.60. Further, there was no main effect of party, $F(1, 200)$ = 0.32, p = .570, η_p^2 < .01. Most importantly, there was no interaction between party and deservingness, $F(1, 200)$ = 0.73, p = .393, η_p^2 < .01.

As we expected, we replicated the results of Ferreira and Botelho (2021) in their first-party conditions, but we also obtained the same results in our added third-party condition. That is, we observed a main effect of (un)deservingness on negative feelings but did not find that (un)deservingness had a stronger impact on first parties than third parties. Study 1b thus conceptually replicates our findings from 1a, demonstrating that another example from the literature purporting to show the effect of deservingness on envy fails the third-party criterion. Furthermore, in Supplemental Study S1, we again replicate this result using a novel vignette.

Finally, we note that we had initially had participants assess others in all conditions rather than asking them to inhabit the first- or third-party roles themselves to avoid self-presentational concerns (Baumeister & Hutton, 1987)—without providing them with a moral justification, people rarely admit to their negative feelings toward others (R. H. Smith & Kim, 2007). However, Study 1b demonstrates that both manipulations of first versus third party (both imagining oneself as the first- or third-party and evaluating the feelings of another person in a first- or third-party role) lead to equivalent results.

Figure 3

Malicious Envy Reported by Participants in Study 1b



Note. In Study 1b, the participants reported malicious envy when the situation was deserved or undeserved and when the participant was in the position of a first or third party. Malicious envy scale goes from 1 (*low*) to 5 (*high*). Error bars display standard error.

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Interim Discussion Studies 1a and 1b: Deservingness Fails the Third-Party Criterion. Are We Mistaking Fairness Concerns for Envy?

In both Studies 1a and 1b (and Supplemental Study S1), we found that deservingness did not pass the third-party criterion: Third parties reported (or were judged to have) as much "malicious envy" as first parties despite no involvement in the scenario themselves and no personal reason to be envious of the advantaged target. That is, in all three of these studies, we see a main effect of (un)deservingness but no interaction between (un)deservingness and party, such that (un)-deservingness increases negative feelings toward the target to the same extent for both potential enviers and nonenviers (third parties).

Importantly, we note that our results in the first-party conditions of Studies 1a, 1b, and Supplemental Study S1 do indeed replicate past results from the literature (that is, more negative feelings based on [un]deservingness), but our full results—with the new inclusion of the third-party conditions—call previous interpretations of such results into question. That is, this well documented variable of "deservingness" that has been argued to be an antecedent of malicious envy does not pass the third-party criterion, suggesting that it is effective at engendering negative feelings toward a target but not envy per se. While it is possible that first and third parties experienced different negative emotions in these studies (e.g., first parties experienced envy and third parties experienced resentment), parsimony militates against this explanation: Both first and third parties showed similar reactions, and thus we have no reason to suspect—and no evidence to support—that these reactions are different. Concern with justice, morality, or fairness is sufficient to explain the negative reactions of both third and first parties; there is no need to conjure an additional envy-driven explanation.

As we have argued, for a variable to be identified as eliciting malicious envy (rather than some other emotion, e.g., indignation at unfairness), it must increase first parties' (i.e., those in a position to experience envy) negative feelings toward the target more than it increases third parties' negative feelings. If a variable simply exerts a main effect on both first and third parties' negative reactions, then there is no way to tell whether this variable is affecting malicious envy. That is, whatever effect the variable appears to have on participants' negative emotions toward the more successful target may instead be driven by some nonenvious emotion or concern.

Indeed, we started this article by noting that part of what makes envy such a dark and perplexing emotion is that it can prompt feelings of ill will toward otherwise innocent people merely by virtue of their success; it is not perplexing (nor does it require envy) to explain why people feel ill will toward bad or undeserving people. The third-party criterion thus provides a simple guidepost for ruling out whether a variable of interest moderates envy.

Study 2: Audience Valuation. Does Audience Admiration Uniquely Fuel Envy?

One worry from Studies 1a and 1b is that participants in the third-party conditions may not have been able to successfully distance themselves from the envy-eliciting situation. Rather than accurately assessing how they or others would feel about an undeserved situation as a third party, perhaps participants simply reported how they would feel were they in the first parties' shoes. If so, this could explain why first- and third-party reactions matched so closely by condition. Therefore, to rule out this possibility, we chose to investigate whether a common, intuitive trigger of envy—the extent to which a relevant audience values the skills and/or traits of a competitor—passes our proposed third-party criterion and thus likely manipulates envy specifically. We chose to investigate this audience valuation variable because being valued highly by others is a sign of social status, and falling behind others in social status can trigger envy (Bolló et al., 2020; Crusius & Lange, 2017). Furthermore, we did not expect this variable to generate negative reactions among third parties: Indeed, people generally wish others who succeed well and having highly valued skills is a sign of competence—a positive trait. Therefore, in line with most commonsense notions of envy, we predicted that manipulating a relevant audience's admiration for a person's competitor would trigger negative feelings based in malicious envy specifically, without triggering the same negative reactions among third parties. We thus predicted that audience valuation would satisfy the third-party criterion. Finding this pattern of results would also demonstrate that participants can inhabit first- or third-party roles in our vignette studies.

Method

We preregistered this study at https://aspredicted.org/W7S_GLT.

Participants

We predetermined to recruit 205 participants from CloudResearch who had 95%+ prior survey approval and were U.S. participants. Participants completed one attention check at the beginning of the survey and were prevented from taking the rest of the survey if they failed this question. At the end of the survey, participants were presented with three comprehension check questions; we planned to exclude participants from analysis if they answered two or more of these questions incorrectly. Of those recruited who completed the survey, 206 participants passed the initial attention check. Of those, 24 failed at least two comprehension checks and were thus excluded from analysis. In total, we analyzed data from 182 participants (92 women, 90 men, $M_{age} = 39.51$ years, $SD = 11.46$) who completed the survey in exchange for $0.45. Sensitivity power analysis suggested this sample size provides 80% power to detect an effect of $\eta_p^2 = .02$.

Experiment Design

We assigned participants to one of four conditions in a 2 (party: first, third) × 2 (audience: values, does not value) between-participants design.

Procedure

Participants read a vignette about two marketing employees at a company, Bob and Garrett. Participants read that lately, Garrett has been excelling at the use of a new marketing software. Across conditions, we varied whether the bosses and other coworkers at the company value one's ability to use the marketing software and thus whether Garrett receives praise for his skills with the software (audience values) or does not (audience does not value). The vignette then stated that Eugene, an accountant at a different company who does not know either Bob or Garrett, hears about

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Garrett's recent success with the new marketing software. For example, in the audience values conditions, participants read,

> Bob and Garrett are two employees who work in the same department at the same company. They both started at the same time and perform the same jobs. Both of their jobs involve marketing.
>
> Garrett has recently been developing his abilities to use company marketing software. He is really excelling at being able to effortlessly navigate the dense software package.
>
> *The bosses of the company, as well as the other employees, care a lot about how well employees can use this software. They think it's the most valuable skill to the company and so Garrett constantly receives praise and accolades from fellow co-workers and bosses for his new skills. Bob always overhears coworkers complimenting Garrett for Garrett's abilities with the software. Bob feels there's every reason to suspect that Garrett will be up for a big promotion soon.*
>
> Eugene is an accountant who works at a different company from Bob and Garrett and does not know either of them. Eugene hears from a friend about how Garrett has been excelling at navigating the new software package at his work.

In the audience does not value conditions, we replaced the paragraph italicized above with the following:

> However, the bosses of the company, as well as the other employees, care little about how well employees can use this software. They don't particularly think it is a valuable skill to the company and so Garrett receives no praise or accolades from fellow co-workers or bosses for his new skills. Bob never hears anyone complimenting Garrett for Garrett's abilities with the software. Bob feels there's no reason at all to suspect that Garrett will be up for a big promotion soon.

After reading the vignette, participants answered four questions assessing either Bob (first-party conditions) or Eugene's (third-party conditions) malicious envy toward Garrett. To increase the generalizability of our envy measurement, these four questions were adapted from the malicious portion of Lange and Crusius (2015a) benign and malicious envy scale. We aggregated these items into a single index ($\alpha = .93$): "How much happier would [Bob/Eugene] be if Garrett were less successful at work?" (1 = *not at all*, 9 = *an extreme amount*), "How much does [Bob/Eugene] hope Garrett will suffer a setback to his success at work?" (1 = *not at all*, 9 = *an extreme amount*), "How frustrated is [Eugene/Bob] by Garrett's success at work?" (1 = *not at all*, 9 = *an extreme amount*), and "How much ill will does [Eugene/Bob] feel toward Garrett?" (1 = *none at all*, 9 = *an extreme amount*). Finally, participants reported their demographic information (age, sex).

Note that, as in Study 1b, we had participants assess others in all conditions (rather than telling them they were in the vignette themselves) to account for self-presentational concerns (Baumeister & Hutton, 1987). Therefore, our "first party" in this case is not the participant but a character who is "first party" to the envy-inducing situation.

Results

Did Audience Valuation Pass the Third-Party Criterion?

Envy. Yes—A 2 (party: first, third) × 2 (audience: values, does not value) ANOVA on target malicious envy ratings revealed an interaction between party and audience, $F(1, 178) = 7.84$, $p = .006$, $\eta_p^2 = .04$ (see Figure 4). First parties were judged to have more malicious envy toward Garrett when the audience valued Garrett's skills with the new marketing software ($M = 4.04$, $SD = 1.89$) than when the audience did not value them ($M = 2.62$, $SD = 1.80$), $t(83.25) = 3.56$, $p = .001$, $d = .77$. Third parties, however, were judged to have similar levels of malicious envy toward Garrett whether the audience valued Garrett's skills ($M = 2.06$, $SD = 1.97$) or not ($M = 2.21$, $SD = 1.87$), $t(93.13) = -0.38$, $p = .702$, $d = -.08$. There was also a main effect of party, such that first parties ($M = 3.38$, $SD = 1.97$) were considered more maliciously envious than third parties ($M = 2.14$, $SD = 1.91$), $F(1, 178) = 19.69$, $p < .001$, $\eta_p^2 = .10$, but this main effect was qualified by the noted interaction above.

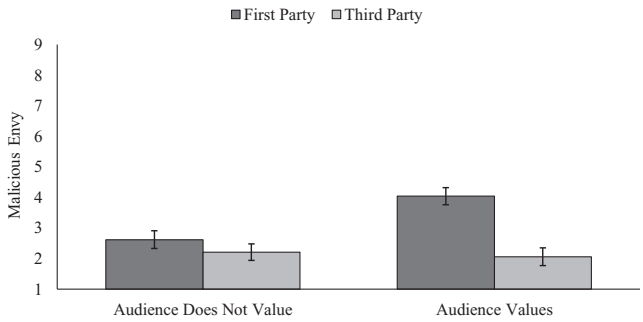
Discussion

As hypothesized, and widely aligned with the literature on social status, first parties were reported to have more negative feelings toward a competitor when that competitor's skills were valued by a relevant audience than when they were not. Crucially, third parties (those not in a position to experience envy) were not reported to harbor more negative feelings toward the competitor when that competitor's skills were valued by the audience. While this finding is perhaps unsurprising by itself, it serves as a sanity check, showing the reader the exact pattern one should expect for a variable that specifically moderates envy. Indeed, these results demonstrate that audience valuation functions as something that uniquely increases malicious envy rather than merely increasing negative reactions from both potential enviers and third parties. That is, we find that audience valuation satisfies the third-party criterion.

In a Supplemental Study, we explored another variable that virtually all theories of envy predict should moderate envy specifically (and one we thought would also pass our third-party criterion), namely the self-relevance of a person's comparison domain (see Supplemental Study S2). Several researchers have argued that as an emotion reliant on social comparison (Alicke & Zell, 2008; Festinger, 1954; Salovey & Rodin, 1984, 1991; R. H. Smith & Kim, 2007; Tesser, 1991), envy should be sensitive to the domain of

Figure 4

Malicious Envy Reported by Participants in Study 2



Note. In Study 2, the participants reported malicious envy when the audience does or does not value the target's coworker and when the participant evaluated the first party or third party's malicious envy. Error bars display standard error.

comparison, and relatedly, the more similar to one's self-identity the target of envy, the more envy one should feel for that target ceteris paribus (Parrott, 1991; Salovey & Rothman, 1991; Schaubroeck & Lam, 2004). In Supplemental Study S1, we manipulated a salient feature of comparison—whether a highly successful worker performs the same job as, or a different job from, the potential envier. As in Study 2, we found that this variable significantly increased attributions of negative feelings for first parties (potential enviers) but did not do so for third parties (nonenviers). Taken together, these studies provide a proof of concept that the third-party criterion can help us identify factors that drive negative feelings in first parties but do not drive them in third parties.

Importantly, these studies also rule out our initial worry from Studies 1a and 1b that participants were unable to distance themselves from the envy-eliciting scenario in our design, causing them to respond as though they were first parties even when asked to imagine themselves (Study 1a) or someone else (Study 1b) in a third-party role. As the results of both Study 2 and Supplemental Study S1 demonstrate, participants are quite capable of taking a third-party perspective and can correspondingly report much lower levels of malicious envy when they do so. Likewise, that third parties were reported to have lower levels of malicious envy in Study 2 and Supplemental Study S1 cannot be explained merely by their being less willing to report negative emotions as a third party: If this were true, then we should not have seen such high levels of emotion (equal to or even surpassing that of first parties) among third parties in Studies 1a and 1b.

Preregistered Mixed-Design Replications

To ensure that the patterns we have observed thus far are not due to participant inability to take a third-party perspective across designs, we ran two preregistered mixed-design replications of Studies 1a and 2 ($N \approx 200$ each; for full methods and tables, see the Supplemental Materials). Across both studies, participants assessed others' feelings rather than their own (as in Studies 1b and 2). And in each of these studies, participants evaluated malicious envy from both a first- and third-party perspective (within subjects). Results from the (un)deservingness replication again showed that undeserved success increased negative reactions overall ($p < .001$) but did not interact with party ($p = .13$), indicating similar emotional reactions among first and third parties (note that the trending interaction of $p = .13$ mirrored the pattern from Figure 1C: When the advantaged target's success was "undeserved," the gap between first and third-party reactions actually narrowed slightly—precisely the opposite pattern expected of a variable that truly moderates envy). In the audience valuation replication, audience praise interacted with party, such that it selectively amplified only first-party envy ($p < .001$), with no effects of question order. Thus, our results demonstrate that participants comfortably differentiate between first- and third-party perspectives. Even when the same participants judge both perspectives, audience valuation but not (un)deservingness passes the third-party criterion.

Study 3: Likeability. Does Disliking Someone Make Us Envious—or Just Hostile?

Having established at least one variable that passes the third-party criterion (audience valuation), we now turn to a further variable we suspect may not—likeability. In the envy literature, researchers have commonly highlighted the role of likeability—the extent to which an envier values or feels close to the envied other—as a potential moderator of benign and malicious envy (e.g., Ferreira & Botelho, 2021; Jankowski & Takahashi, 2014; Lange & Crusius, 2015b): The less an envier likes someone, the more malicious envy they feel, and the more an envier likes someone, the less malicious envy they feel in situations of upward social comparison. On the dual account of envy, this effect follows directly from appraisal patterns: Liking augments appraisals of high deservingness, while disliking augments appraisals of low deservingness (Crusius & Lange, 2017; Feather, 1992, 1999). However, as with deservingness, manipulations of likeability may tap psychologies other than envy (e.g., hatred), and thus what looks to be an increase in "malicious envy" toward a target may instead reflect merely an increase in broader negative feelings toward that target. To test this possibility, we provided participants with a scenario containing information that could plausibly spark malicious envy—a neighbor is doing very well at work—but across conditions we manipulated the cruelty of this neighbor. We hypothesized that by manipulating the neighbor's cruelty, we could make participants dislike the neighbor and that this dislike would lead to increased "malicious envy"—but these broad negative feelings would be felt by both potential enviers and by third parties. Thus, we predicted that likability would fail the third-party criterion.

Method

We preregistered this study at https://aspredicted.org/1TF_Q8B.

Participants

We predetermined to recruit 205 U.S. participants from CloudResearch who had 95%+ prior survey approval and were U.S. participants. Participants completed one attention check at the beginning of the survey and were prevented from taking the rest of the survey if they failed this question. At the end of the survey, participants were presented with three comprehension check questions; we planned to exclude participants from analysis if they answered two or more of these questions incorrectly or failed a final attention check. Of those recruited who completed the survey, 204 participants passed the initial attention check. Of those, 26 failed at least two comprehension checks or a final attention check and were thus excluded from analysis. In total, we analyzed data from 178 participants (74 women, 104 men, M_{age} = 39.40 years, $SD = 10.55$) who completed the survey in exchange for $0.50. Sensitivity power analysis suggested this sample size provides 80% power to detect an effect of $\eta_p^2 = .02$.

Experiment Design

We assigned participants to one of four conditions in a 2 (party: first, third) × 2 (likeability: likable, unlikable) between-participants design.

Procedure

Participants read a vignette describing a neighbor who has recently been doing well at work. In the first-party condition, participants were told to imagine this was their neighbor, while in

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the third-party condition, participants were told to imagine this was Bob's neighbor. Across conditions, we varied how we described the neighbor. In the likable condition, the neighbor was described as helpful and good natured, while in the unlikable condition, the neighbor was described as rude and needlessly cruel. For example, in the first party, unlikable condition, participants read,

> You are out mowing your lawn one day when you observe your neighbor, Greg, out walking his dog. As a salesman, Greg is always pestering people to buy things from him even though they tell him they don't want anything and to please leave them alone. Apparently Greg has been doing quite well at his sales job lately.
>
> You see Greg cursing at his dog while yanking on its leash. Greg gives the dog a kick, and it lets out a crying whimper. You notice some garbage lying in the street. You recall that Greg finds it funny to kick over people's garbage cans and spread trash over their yards.

And in the third-party, likable condition, participants read,

> Bob is out mowing his lawn one day when he observes his neighbor, Greg, out walking his dog. As a salesman, Greg is always bringing Bob interesting new things to buy. Apparently Greg has been doing quite well at his sales job lately.
>
> Bob sees Greg petting his dog as he takes it for a walk. Greg whistles, and the dog runs right up to him. Bob notices some garbage lying in the street. Bob recalls that Greg often spends his time picking up garbage in the neighborhood and removing it from people's yards.

After reading their assigned version of the vignette, participants then answered six questions designed to assess their malicious envy for the neighbor, Greg. To further increase generalizability, we combined all of our measures from the previous studies: Two of these questions were adapted from van de Ven et al. (2012) and Ferreira and Botelho (2021): "Would you secretly wish for Greg to lose sales clients at work?" and "Would you gossip about Greg to others?" (1 = *not at all*, 8 = *very much so*). The four additional measures were the same as those used in Study 2 adapted from Lange and Crusius (2015a): "How much happier would you be if Greg were less successful at work?," "How much do you hope Greg will suffer a setback to his success at work?," "How frustrated are you by Greg's recent success at work?" (1 = *not at all*, 8 = *an extreme amount*), and "How much ill will do you feel toward Greg?" (1 = *none at all*, 8 = *an extreme amount*). All six measures showed a high degree of consistency ($\alpha = .95$). As preregistered, given a Cronbach's $\alpha > .6$, all six malicious envy items were combined into a single composite measure.

As a manipulation check, we also measured the perceived likeability of the neighbor: "How much do you/does Bob like Greg?" (1 = *strongly dislike[s] him*, 9 = *strongly like[s] him*). The manipulation was successful: Participants liked (or thought Bob liked) the neighbor more when the neighbor was described as likable ($M = 6.85$, $SD = 1.38$) than when he was described as unlikable ($M = 2.42$, $SD = 2.04$), $t(154.54) = 17.02$, $p < .001$, $d = 2.45$. Finally, participants reported their demographic information (age, sex).

Results

Did Likeability Pass the Third-Party Criterion?

Envy. No—a 2 (party: first, third) × 2 (likeability: likable, unlikable) ANOVA on target malicious envy ratings again revealed no interaction between party and likeability, $F(1, 174) = 0.10$, $p = .751$, $\eta_p^2 < .01$. As predicted, there was only a main effect of likeability, $F(1, 174) = 131.84$, $p < .001$, $\eta_p^2 = .43$ (see Figure 5), such that participants expressed more malicious envy (or evaluated others as more maliciously envious) when the neighbor was unlikable ($M = 4.98$, $SD = 1.79$) than when the neighbor was likable ($M = 1.98$, $SD = 1.67$), $t(175.02) = 11.55$, $p < .001$, $d = 1.73$. Importantly, in the unlikable conditions, there was no statistical difference between the malicious envy ratings for first parties ($M = 4.93$, $SD = 2.00$) and third parties ($M = 5.03$, $SD = 1.58$), $t(83.39) = -0.27$, $p = .785$, $d = -.06$. The same held true in the likable conditions: Participants judged themselves as first parties ($M = 2.01$, $SD = 1.72$), and others as third parties, ($M = 1.95$, $SD = 1.63$) as similarly maliciously envious, $t(86.98) = 0.17$, $p = .863$, $d = .04$.

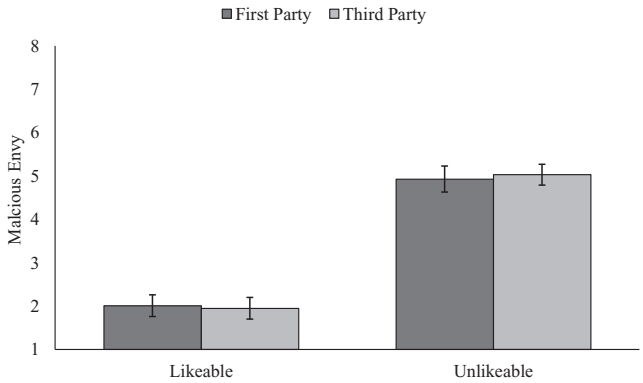
Discussion

Study 3 demonstrates another variable that does not pass the third-party criterion: Decreasing an envied target's likability increases both first- and third-party negative feelings toward the target. Rather than envy, it is likely that turning down likeability can activate broad negative feelings for which more specific social comparison-based explanations rooted in envy are extraneous. The results of this study suggest that (dis)likability can certainly increase negative feelings, but that it does so for both enviers and nonenviers (i.e., third parties).

While these results suggest that likeability in this context did not drive feelings of envy, the relationship between envy and likeability is likely to be complex. For example, envy may cause us to dislike those we already envy for other reasons—perhaps as a way to justify our aggression toward them (Moncrieff & Lienard, 2024). Whatever likeability's relationship to envy, it is crucial that envy researchers ensure that when manipulating a variable like likeability that they are in fact manipulating the emotion they wish to measure. The third-party criterion provides a useful heuristic for evaluating whether said manipulation truly operates on envy.

Figure 5

Malicious Envy Reported by Participants in Study 3



Note. In Study 3, the participants reported malicious envy when the neighbor was likable or unlikable and when participants rated their own (first party) malicious envy or a third party's malicious envy. Malicious envy scale goes from 1 (*low*) to 8 (*high*). Error bars display standard error.

General Discussion

As a competitive emotion activated by threats to one's status (Feather & Nairn, 2005; Fiske, 2011), envy is not an emotion we would expect third parties to experience—disconnected as they are from any competition with the enviable target. Our proposed third-party criterion thus offers a simple but powerful tool for determining if something is not an effective manipulation of envy: If a variable increases negative reactions similarly among potential enviers (i.e., first parties) and uninvolved observers (i.e., third parties), then it is probably not manipulating envy.

In our studies, both audience valuation (Study 2 and Supplemental Study S4) and the self-relevance of one's comparison domain (Supplemental Study S2) passed our third-party criterion, increasing envy among potential enviers but not third parties, thereby qualifying as candidate moderators of envy. By contrast, (un)deservingness and likeability did not: While both manipulations increased negative feelings across conditions (replicating past results), neither uniquely augmented negative feelings for first parties above those of third parties (Studies 1a, 1b, 3; Supplemental Studies S1 and S3). Therefore, these manipulations failed the third-party criterion: They do not affect envy specifically but instead appear to elicit broader social or moral emotions. Although previous research has interpreted increases in negative reactions to an undeserved and unlikable target's success as malicious envy, our results call this interpretation into question. Rather, our results indicate that neither (un)deservingness nor likeability are likely to be robust moderators of envy per se, at least not in the vignette designs that have typically been used by past literature.

This conclusion is particularly noteworthy in the case of deservingness. Though we stake no claim in the "dual" versus "unitary" envy debate (see Crusius et al., 2020), the fact that (un)deservingness fails the third-party criterion undermines the dual account's claim that this variable cleanly distinguishes between benign and malicious envy. As we have noted, one plausible reason for this failure is that (un)deservingness manipulations may elicit responses grounded in outrage, fairness, or moral judgment—emotional reactions that affect third parties as much as first parties. A related possibility is that (un)deservingness manipulations may activate resentment rather than envy (R. H. Smith & Kim, 2007). Though emotion researchers disagree on the exact dividing line between resentment and envy (see e.g., Crusius et al., 2020; Feather & Nairn, 2005; Feather & Sherman, 2002; Silver & Sabini, 1978; R. H. Smith, 1991; R. H. Smith et al., 1994, 1996; R. H. Smith & Kim, 2007), Feather and Sherman (2002) found that participants' resentment and envy loaded onto two distinct factors: "Resentful" loaded onto a factor along with "angry," "indignant," and "a feeling of injustice," while "envious" loaded onto a factor with "jealous" and "be like the stimulus person [the advantaged target]." If resentment is triggered specifically by feelings of anger at unfairness or injustice—emotions experienced by third parties as much as first parties—then resentment represents a strong candidate for what (un)deservingness manipulations are actually manipulating. Our findings thus highlight the value of the third-party criterion not only as a methodological test for isolating envy, but also as a broader tool for clarifying emotional boundaries: There are many reasons to wish others ill (some perhaps even noble), but only a small portion of these negative feelings constitutes envy.

More generally, our findings underscore the importance of accurate envy *measurement*—that is, in determining the most precise and comprehensive ways to measure an emotion like envy. In this respect, we echo Cohen-Charash and Larson's (2017) call for more varied and nuanced approaches to envy measurement. A related (but different) approach that we have presented here is to include appropriate negative controls in which one should not see an effect (i.e., variables that pass the third-party criterion): It is not enough to show that a variable produces an effect when it should; it must also *not* produce an effect when it should not. As we see with variables that fail the third-party criterion, many different emotions can share behavioral and cognitive outputs. For instance, envy, hatred, concerns with fairness, and morality can all lead to derogation and dislike of an advantaged target despite otherwise distinct elicitors, cognition, motivations, behavior, and terminating conditions (Sell et al., 2021). Thus, merely measuring *outputs* of envy risks measuring other emotions whose outputs can share envy's outputs (e.g., dislike, hatred, anger, unfairness, and resentment), and what may seem like successful manipulations of envy may instead be successful manipulations of other emotions. Failing to accurately measure the emotions we wish to manipulate can lead to confusion about the nature of the emotions themselves, their effects, and the factors that influence them that compound and proliferate in the literature. Arguably, this has already happened, both with respect to envy and other emotions as well (Shaw & Choshen-Hillel, 2017; Sznycer et al., 2017). As with many a problem, the first step to a solution is identifying it: In that spirit, we hope the present work serves as a diagnostic tool—one that helps researchers in the envy literature spot potential misattributions and move the field closer to conceptual clarity.

Do our results mean that deservingness and likeability have no meaningful connection to envy? Not necessarily. Imagine, for example, a very productive employee who is promoted despite always being a bit late to meetings. A third party might agree that this employee deserves her promotion and so feel no ill will toward her. However, a work competitor may look at the same facts and zero in on his coworker's tardiness (and wonder, "does this employee really deserve her promotion if she's always late to meetings when I'm always on time?"). Without some initial spark of envy, this variable would not matter (as revealed by the hypothetical third party response), but with that initial spark, the "lack of deservingness" may fuel further envy. In this way, appraisals of deservingness and/or likeability may be unrelated to the elicitation of envy in the first place; instead, perceptions of unfairness and dislike may be outputs of an already envious mind, ones designed to motivate derogation of the envied target and other, more malicious behavioral responses (Hill & Buss, 2006).

Correspondingly, other researchers have noted that factors like deservingness might serve as justifications for our negative feelings rather than being the root of them (e.g., see Cohen-Charash, 2009; Moncrieff & Lienard, 2024; R. H. Smith & Kim, 2007). These justifications can then intensify the negative feelings toward the envied target. This represents an intriguing way in which appraisals like deservingness may be related to envy and feed into subsequent envy reactions. Thus, we do not wish to rule out that these variables could matter more to enviers than to third parties but argue that it is vitally important to demonstrate precisely how they do. Our results make clear that envy researchers should proceed with caution when

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designing unambiguously undeserving scenarios, because if they manipulate deservingness in a way that leads most people (even third parties) to see the advantaged target's success as undeserved, then the negative feelings they elicit may be driven by concerns with fairness rather than envy.

Our findings also raise the possibility that other variables thought to affect envy might fail the third-party criterion. One such candidate is hubristic pride (Tracy & Robins, 2007b). Attribution researchers have argued that pride contains two distinct dimensions: "authentic" pride and "hubristic" pride. Authentic pride occurs when we attribute events consistent with our identity goals to internal, unstable, and controllable causes ("I aced the test because I studied hard"), whereas hubristic pride occurs when we attribute those events to internal, stable, and uncontrollable attributions ("I aced the test because I'm a genius"; Tracy & Robins, 2007a, 2007b). Given that expressions of pride often accompany accomplishment and can signal increases in one's status (Shariff et al., 2012), envy researchers have argued that pride displays augment envy (Lange & Crusius, 2015b). In particular, Lange and Crusius (2015b) have argued—and claim to have demonstrated empirically—that authentic pride preferentially activates benign envy, whereas hubristic pride preferentially activates malicious envy. However, in line with our results here, it may be that displays of hubristic pride are perceived as less deserved and/or likable than more authentic displays (Feather, 1999). If the same reasoning from Studies 2–3 holds, third parties may express as much negative feelings toward hubristic pride displays as first parties do, suggesting we cannot assume that greater negative feelings toward a displayer of hubristic pride constitutes greater malicious envy toward that displayer. As work on this question and others progresses, we believe the third-party criterion will serve as a useful tool for adjudicating between elicitors of envy and nonenvious emotions.

Beyond its theoretical and methodological contributions, the third-party criterion also holds practical value. Any interventions based on findings from the empirical literature will benefit from being able to differentiate envy from other emotions. For example, if people's negative feeling toward an actor are based on concerns with morality or justice, then interventions designed to reduce competitiveness or social comparison will be ineffective at quelling these negative feelings. In workplaces or schools, distinguishing between envy and other negative reactions (e.g., moral outrage or dislike) can help leaders respond more appropriately to interpersonal conflict. Rather than assuming competition always drives envy-based negativity toward more successful peers, practitioners might consider whether a situation instead reflects concerns about fairness, recognition, or social belonging. The third-party criterion offers a practical heuristic: If third parties would plausibly feel just as negative toward a successful individual as first parties do, it may suggest that the response stems from non-envy-based concerns. Likewise, in clinical setting, therapists working with clients harboring intense negative feelings toward others might benefit from understanding whether those feelings stem from envy-driven competitiveness and social comparison or from broader moral evaluations. Depending on the underlying emotion, effective interventions may target comparative thinking (e.g., in the case of envy) rather than perceptions of unfairness or injustice (in the case of resentment). By offering a tool to clarify the emotional process at play, the third-party criterion may aid efforts to diagnose and defuse status-related conflict more effectively.

Constraints on Generality

As mentioned at the outset, it is important to note that our studies examined moderators of malicious envy only, without probing reactions to benign envy. We made this decision because there is considerable debate in the literature about whether envy truly has distinct benign and malicious subtypes (for an overview, see Crusius et al., 2020). Furthermore, malicious envy fits people's intuitive definition of envy and has even been referred to as "envy proper" by some envy researchers (e.g., Parrott, 1991; R. H. Smith & Kim, 2007). However, the decision to focus on malicious envy means that we provided no evidence for how this procedure works for reactions based in benign envy. Had we measured benign rather than malicious envy, it is possible that the same variables that passed the third-party criterion would still have passed and that the same variables that failed would still have failed. For instance, hearing about a likable, deserving person's recent success may inspire both third parties and potential enviers to the same extent (inspiration being one common measure of benign envy). These possibilities are ultimately empirical questions for future research to investigate.

One additional limitation worth highlighting is the fact that we had little information about the demographic breadth of our samples. Because we did not have a specific theoretical rationale or any hypotheses regarding demographic variables (e.g., socioeconomic status and ethnicity) or personality traits, we collected only basic demographic data (age and gender). Although CloudResearch and Prolific typically provide relatively diverse samples (Douglas et al., 2023), we acknowledge that lacking detailed demographic characterization may limit the generalizability of our findings to broader populations. Future research that explicitly samples across varied demographic dimensions could provide further replications of our work and external validation of the third-party criterion.

Further, we should mention that the current research—because it sought to replicate past work on envy—involved only vignette-based designs meant to tap how people feel about others' success. While some work has shown that results from carefully constructed vignette-based designs can mirror those from more direct, behavioral experiments (e.g., Hainmueller et al., 2015), we believe it would be valuable to measure participants' actual behavior rather than just their self-report of their emotions or speculations about others' emotions. Such work will be informative about not just whether people feel negative about others' success, but also whether they will take active steps to harm or undermine successful others. Understanding people's attitudes is important, but it is equally important to know how often these attitudes manifest as behaviors in the real world.

Indeed, while deservingness and likeability failed the third-party criterion when replicating these vignette-based designs meant to assess people's attitudes, it is possible that they would pass given designs that utilize more direct behavioral measures (e.g., designs that assess one's willingness to pay costs to inhibit another's progress). We know that in some situations, first parties are more willing to pay costs to punish others than third parties, even though both will say the action is wrong and agree that the action should be punished (Pedersen et al., 2013). This suggests that when responding to undeserved situations, third parties may indeed feel negative about others' immoral or undeserved success but might be much less willing to pay costs to right those wrongs than actual victims. If this is the case, then behavioral experimental designs may

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yield different results than those observed here, and there might be cases where deservingness exerts a stronger impact on first than third parties. While we would expect for first parties to be more willing to put their proverbial money where their mouths are and pay costs to reduce the success of competitors generally (e.g., first parties will even do so when others receive more in an economic game because of random chance, Dawes et al., 2007), the more relevant question is whether deservingness would exert a stronger impact on first parties than third parties over and above what would otherwise be expected absent information about the deservingness of the situation.

Regardless of how that experiment turned out, given that the evidence for the effects of appraisals of deservingness and likeability on envy (and thus the dual account of envy itself) rests on results obtained from the vignette-based experimental designs examined here, the current results call the interpretation of that evidence into question: In short, the present work suggests that some of the key manipulations thought to promote malicious envy may instead prompt more general negative feelings, and it is these negative feelings—rather than envy per se—that can explain why we begrudge others' success.

Conclusion

As Adam Smith argued in *The Theory of Moral Sentiments*, "Envy is that passion which views with malignant dislike the superiority of those who are really entitled to all the superiority they possess" (A. Smith, 1853, p. 358). Aligned with Adam Smith, we find that disliking those who are undeserving of their advantages need not be explained by envy (e.g., caring about fairness can explain that). Rather, envy may be most strongly directed toward those whose advantages are genuinely earned—and perhaps precisely because they are.

Envy is a complicated emotion. To the extent that envy is a unique emotion unto itself, understanding it requires probing how and why it is separable from other concerns, preferences, and emotions that are in its orbit: for example, anger, resentment, injustice, desire, hatred, jealousy, and so forth. Lacking clarity on such matters can lead to severe consequences, since theories of emotions inform what interventions aim to achieve. Indeed, without a clear understanding of emotions like envy and the specific factors that influence them, the inferences we draw from our research will be incomplete or even incorrect. In the same way that all that glitters is not gold, all hatred aimed at successful others need not be rooted in envy. We hope the current research provides a useful tool for telling the difference between negative feelings rooted in envy and those rooted in other concerns—for determining when what looks like the green-eyed monster turns out only to be a mirage.

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